Free-for preservation under substitution

The Open Logic Project contributors

openlogic:first-order-logic/axiomatic-deduction:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-first-order-logic-axiomatic-deduction/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: standalone; consumed by lem:ChangeBdVar (this batch).

Editorial note. The label lem:free-for names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Lemma 0.1. *If x is free for c in A and y is not free in A, then x is free for y in $A[y/c]$.*

Proof. If x is not free for y in $A[y/x]$, then some free occurrence of y in $A[y/c]$ falls within the scope of a quantifier $\forall x$. But y is not free in A, by hypothesis, so all such occurrences come from the substitution $[y/c]$. So some occurrence of c falls within the scope of $\forall x$ and x is not free for c in A. □